

DATA VORTEX A'26 — ROUND 2

Technical Report — Rebuilding the Social Engine

Task: NLP-based semantic classification of labelled social-media text

Dataset: Labeled_Social_NLP_Training_Data.csv

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1. Problem Definition

The Round 2 task focuses on rebuilding a semantic comprehension layer for social-media text. The supplied dataset contains labelled textual posts with two prediction targets: sentiment_label and topic_category.

The objective is to develop NLP classification models capable of extracting semantic information from short, noisy social-media text and assigning the appropriate sentiment and topic labels.

Sentiment classes: Negative, Neutral, Positive.

Topic classes: Account_Security, Community_Discussion, Feature_Feedback, Technical_Issues.

2. Dataset Understanding

Property	Observed Value
Total rows	9,000
Columns	4
Missing values	None
Duplicate rows	None
Unique text IDs	9,000
Rows belonging to duplicated texts	2,087
Conflicting duplicate sentiment labels	0
Conflicting duplicate topic labels	0

Sentiment distribution: The three sentiment classes are perfectly balanced, with 3,000 samples each.

Topic distribution:

Topic	Count	Percentage
Community_Discussion	7,752	86.13%
Technical_Issues	815	9.06%
Feature_Feedback	297	3.30%
Account_Security	136	1.51%

The topic target is highly imbalanced. This imbalance was explicitly considered during model development and evaluation.

3. Preprocessing Pipeline

The preprocessing strategy was deliberately conservative so that semantic information and social-media signals were not unnecessarily removed.

Step	Processing
1	Load the original labelled CSV dataset.
2	Validate missing values, duplicates and label consistency.
3	Normalize leading/trailing whitespace in post_text.
4	Remove empty text records; none were removed.
5	Create a group-based train/test split using post_text.
6	Verify zero exact post_text overlap between train and test.
7	Apply TF-IDF feature extraction for the classical topic model.
8	Tokenize text using DistilBERT tokenizer for sentiment modelling.

A GroupShuffleSplit strategy was used with post_text as the grouping variable. The final split contains 7,178 training rows and 1,822 testing rows. There is zero exact text overlap between the two sets, reducing the risk of duplicate-text leakage.

4. Model Selection

Several classical machine-learning baselines were evaluated using TF-IDF representations. Logistic Regression, Linear SVM and Naive Bayes were tested for both sentiment and topic classification.

Task	Model	Accuracy	Macro F1
Sentiment	Logistic Regression	58.23%	0.5833
Sentiment	Linear SVM	55.65%	0.5574
Sentiment	Naive Bayes	57.03%	0.5689
Topic	Logistic Regression	88.20%	0.3413
Topic	Linear SVM	91.66%	0.5465
Topic	Naive Bayes	85.51%	0.2305

An enhanced TF-IDF representation combining word n-grams and character n-grams improved topic performance substantially. Class weighting was used for the imbalanced topic target.

Final sentiment model: DistilBERT (distilbert-base-uncased), fine-tuned for three sentiment classes.

Final topic model: Linear SVM with C=2.0, balanced class weights, combined word and character TF-IDF features.

5. Training Methodology

For sentiment classification, DistilBERT was fine-tuned using the 7,178 training samples. Text was tokenized with a maximum sequence length of 128 tokens. Training used two epochs, a learning rate of 2×10^{-5} , batch size of 8 and weight decay of 0.01.

The model was evaluated after training using the held-out test set. The best model was selected using Macro F1 as the evaluation criterion.

For topic classification, word-level TF-IDF features used 1–3 grams with a maximum of 100,000 features. Character-level TF-IDF features used 3–6 character grams with a maximum of 100,000 features. The two sparse matrices were combined and supplied to a Linear SVM with C=2.0 and class_weight='balanced'.

The use of character n-grams helps represent variations and partial patterns in short social-media text, while word n-grams capture lexical context.

6. Evaluation Metrics

The evaluation uses accuracy, precision, recall, F1-score and confusion matrices. Macro-averaged metrics are particularly important for topic classification because the topic labels are highly imbalanced.

Task	Model	Accuracy	Macro F1	Weighted F1
Sentiment	DistilBERT	68.33%	0.6781	0.6777
Topic	TF-IDF + Linear SVM	93.08%	0.6160	0.9157

The sentiment model achieved 68.33% accuracy and a Macro F1 of 0.6781. The topic model achieved 93.08% accuracy and a Macro F1 of 0.6160. The difference between topic accuracy and Macro F1 reflects the strong topic-class imbalance.

7. Confusion Matrix — Sentiment

Actual \ Predicted	Negative	Neutral	Positive
Negative	459	70	67
Neutral	128	312	171
Positive	52	89	474

The model correctly classified 459 Negative, 312 Neutral and 474 Positive test samples. Neutral has the weakest recall at 51.06%. The largest individual confusion is Neutral predicted as Positive, occurring in 171 cases.

8. Confusion Matrix — Topic

Actual \ Predicted	Account_Security	Community_Discussion	Feature_Feedback	Technical_Issues
Account_Security	8	15	2	0
Community_Discussion	0	1558	0	0
Feature_Feedback	0	53	7	0
Technical_Issues	0	56	0	123

Community_Discussion is classified very strongly, with all 1,558 test samples correctly identified. The minority classes are harder to detect. Feature_Feedback has recall of 11.67%, while Account_Security has recall of 32.00%. Many minority-class samples are classified as Community_Discussion.

9. Error Analysis

The error patterns reveal different challenges for the two NLP tasks.

Sentiment classification:

- Neutral language is the most difficult sentiment class, with recall of 0.5106.
- Neutral → Positive is the largest confusion pair with 171 samples.
- Neutral → Negative accounts for another 128 misclassifications.
- The relatively balanced class distribution means these errors are not primarily caused by class-frequency imbalance.

Topic classification:

- The topic distribution is highly skewed toward Community_Discussion (86.13%).
- Feature_Feedback has only 60 test samples and recall of 0.1167.
- Account_Security has only 25 test samples and recall of 0.3200.
- Feature_Feedback and Account_Security samples are frequently absorbed into the dominant Community_Discussion class.
- Accuracy alone therefore gives an incomplete picture of topic performance; Macro F1 provides a more balanced assessment across classes.

10. Data Quality and Leakage Control

Dataset validation found no missing values, no duplicate rows and no duplicate text IDs. Although repeated post_text values exist, duplicate texts had consistent sentiment and topic labels.

To prevent repeated text from appearing across training and testing, the train/test split was grouped by post_text. The final split has zero exact text overlap between training and testing data.

11. Model Comparison and Justification

Task	Approach	Accuracy	Macro F1
Sentiment	TF-IDF + Logistic Regression	59.66%	0.5973
Sentiment	DistilBERT	68.33%	0.6781
Topic	Baseline TF-IDF + Linear SVM	91.66%	0.5465
Topic	Final TF-IDF + Linear SVM	93.08%	0.6160

DistilBERT was selected for sentiment because the fine-tuned transformer produced stronger held-out sentiment metrics than the tested classical alternatives. For topic classification, the combined word/character TF-IDF representation with a balanced Linear SVM provided a practical and computationally efficient solution while improving Macro F1 over the baseline.

12. Limitations and Future Improvements

- The topic dataset is strongly imbalanced, particularly for Account_Security and Feature_Feedback.
- Minority-topic recall remains an important area for improvement.
- Sentiment errors show difficulty distinguishing Neutral language from positive and negative language.
- Further improvements could investigate targeted data augmentation, domain-specific pretrained language models, threshold/calibration strategies and additional minority-class examples.

13. Conclusion

The Round 2 NLP pipeline combines leakage-aware dataset preparation, classical TF-IDF modelling and transformer-based semantic modelling. The final sentiment model achieved 68.33% accuracy with a Macro F1 of 0.6781, while the final topic model achieved 93.08% accuracy with a Macro F1 of 0.6160.

The evaluation demonstrates the importance of examining per-class metrics and confusion matrices rather than relying on accuracy alone, particularly for the highly imbalanced topic classification task.